

# Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

**Benchmark:** ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

This is the most critical structural mismatch for this deployment. The label schema is exclusively single-correct-answer MCQ with 2–5 candidate options [Q27], and every sampled example confirms this format [DATASET-D1, all]. The deployment explicitly requires multi-perspective acknowledgment of contested historical and civic questions [elicitation A3], but the benchmark's output taxonomy structurally cannot represent or score such pluralistic responses. The four prompt-configuration variants tested vary only in prompt language and output script [Q47], not in label structure. Contested questions like King Abdullah I and Zionism [DATASET-D1], GCC founding states [DATASET-D25], and 'elements of a state' with 'occupation' as a distractor option [DATASET-D18] are all reduced to single correct labels reflecting one curriculum's framing. The future-work note proposing short-answer and essay questions [Q88] tacitly acknowledges this limitation. Open-ended Arabic evaluation alternatives exist (AraGen, 'Beyond MCQ' — [WEB-5, WEB-6]) but are not part of ArabicMMLU itself.

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27   | "Each question has 2–5 candidate answers, with one correct answer." (p.3)                                                                                                                                                                                                                                                                                                                            |
| Q47   | "We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)                                                                                                           |
| Q88   | "For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)                                                                                                                                                                              |
| D1    | دحو مامتإ: ني طسلف يف اهعماطأو ةي نوي هصلا ةك رحلا نم لوألا هلل ادبع كل مل فق اوام نم تسيل ةيتآل نم ةدحاو هضفر / ني طسلف ةمأل قودنص معد لإ برعل ةودع / قاربلا طئاح لإ ارهم دوهيلا اعطاع هضفر / م 1950 ماغ ني تفضللا اعطاق اضفر روفل ب دعوو وكيب سكي اس ةدهامع — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D18   | — Palestine MS geography: elements of a state — عداي سلا / لال تحال / مي لقال / بعشلا: ادع ةلودل رصانع نم يلي ام عجم "occupation" appears as a distractors option                                                                                                                                                                                                                                    |
| D25   | — Jordan civics HS: false claim about GCC founding states — correct answer is "No" since Yemen and Iraq were not founders هدمتملا هي برعل تارامال) هي جيجلخل نواعتلا سلجم ةماقلا لتعس تي لتل لودلا                                                                                                                                                                                                   |
| WEB-5 | 'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF ( <a href="https://arxiv.org/abs/2510.24328">https://arxiv.org/abs/2510.24328</a> )                                                                                                                                                                                                                                      |
| WEB-6 | AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation ( <a href="https://arxiv.org/abs/2510.13430">https://arxiv.org/abs/2510.13430</a> )                                                                                                                                                                                                                                              |

## Strengths

- MCQ format produces well-defined, reproducible scoring for factual recall tasks [Q27]
- Multiple prompt-configuration variants enable analysis of prompt-language sensitivity [Q47]

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|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Output label categories are uniform single-correct MCQ [Q27], reviewed against deployment requirement for multi-perspective acknowledgment [elicitation A3] — fundamental misalignment.

**OO-2:** Missing categories: multi-perspective/multi-correct labels, perspective-flagging labels, explanatory/short-answer outputs, abstention labels for contested content. None present in benchmark [Q27, Q88].

**OO-3:** The single-correct-answer assumption encodes a 'one curriculum framing is correct' value that conflicts with the deployment's pluralistic requirement [elicitation A3]; instances like the King Abdullah I question [DATASET-D1] and 'occupation' distractor [DATASET-D18] make this concrete.

**OO-4:** Stakeholder-driven taxonomy redesign needed for deployment use; supplementary frameworks (AraGen 3C3H, 'Beyond MCQ') identified but not specifically targeted at contested historical content [WEB-5, WEB-6, net\_new\_findings].

**OO-5:** Documented: structural validity violation (construct of multi-perspective knowledge cannot be represented), content validity violation (multi-perspective categories absent), and external validity risk (benchmark performance will not predict pluralistic response quality).

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| D1    | دحو ومامتإ: نني طسلف في اهعاطأو ةينوي هصل ةكرحل نم لوأل اهلل ادبع كل مل افقاوم نم تسيل ةيتأل نم ةدحاو هضفر / نني طسلف ل ةمأل قودنص معد ل برعلا ةودع / قارب ال طئاح ل ارم دوهي ل ااطع هضفر / ماع نني تفضل ااعطا ق اصف روفل ب دعوو وكيب سكي اس ةدهاعم — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective |
| D18   | دعاي سل / لال تحال / مي ل قال / بعش ل: ادع ةلودل رصانع نم يلي ام عجم — Palestine MS geography: elements of a state — "occupation" appears as a distractors option                                                                                                                                                                                                                          |
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## Requires Expert Verification

- Which specific contested questions in the full dataset would require multi-perspective treatment under the deployment's pluralistic framing

## Remediation

**Gap 1: Single-correct-answer MCQ format cannot represent multi-perspective acknowledgment required by deployment [Q27, elicitation A3]**

**Recommendation:** Supplement ArabicMMLU with an open-ended Arabic generation evaluation (e.g., AraGen 3C3H [WEB-6] or 'Beyond MCQ' [WEB-5]) and develop a custom rubric that scores perspective-flagging on contested historical/civic items, with explicit credit for acknowledging country-specific framings.

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